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Subject:- Operating System**

Unit I

Evolution of operating systems, Types of operating systems. The process concept, system programmer's view of processes, operating system's views of processes, operating system services for process management

Introduction

An **operating system** (OS) is a software program that serves as a conduit between computer hardware and the user. It is a piece of software that coordinates the execution of application programs, software resources, and computer hardware. It also aids in the control of software and hardware resources such as file management, memory management, input/output, and a variety of peripheral devices such as a disc drive, printers, and so on. Every computer system must have at least one operating system to run other applications. Browsers, MS Office, Notepad Games, and other applications require an environment to execute and fulfill their functions. This blog explains the evolution of operating systems over the past years.

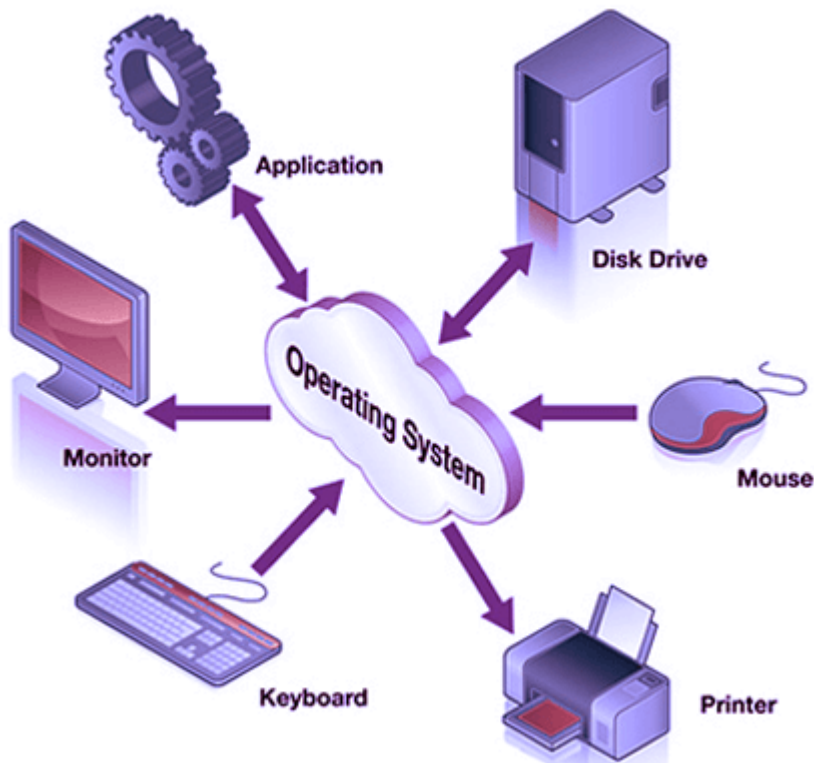
What do you mean by Operating system?

No matter its size and application, every computer needs an operating system to make it functional and useful. The operating system is an integral part of modern computer systems. It is a well-organized collection of programs that manages the hardware.

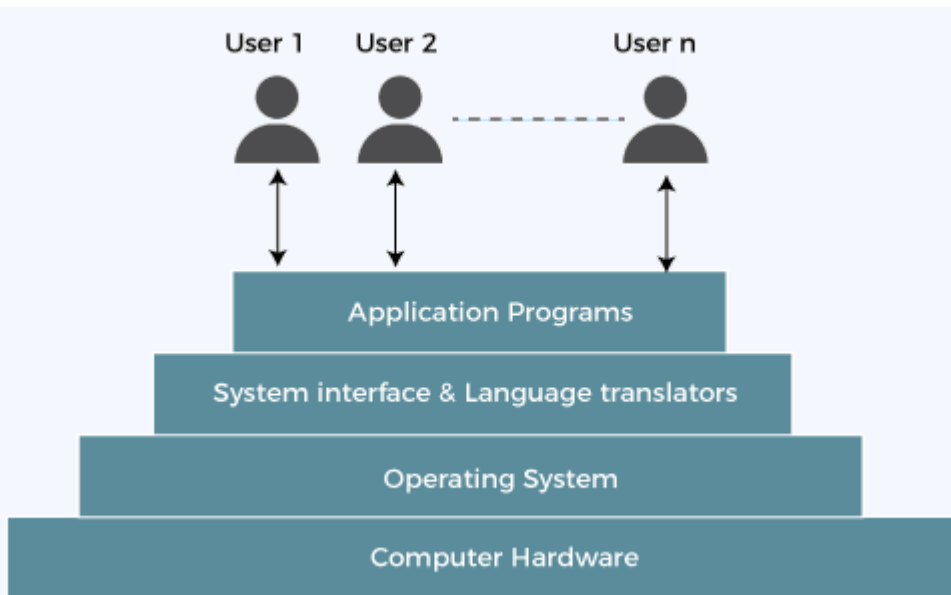
An Operating System provides an interaction between the users and computer hardware. A user is a person sitting at the computer terminal concerned about the application rather than the architecture of the computer. The user never interacts with the hardware directly. To get the services of the hardware, he has to request through the operating system.



The operating system is a primary resource manager. It manages the hardware, including processors, memory, Input-Output devices, and communication devices.



The operating system operates either in kernel mode or user mode. Compilers and editors run in user mode, whereas operating system code runs in kernel mode.



What is Evolution of Operating Systems?

Operating systems have progressed from slow and expensive systems to today's technology, which has exponentially increased computing power at comparatively modest costs. So let's have a detailed look at the evolution of operating systems.

The operating system can be classified into four generations, as follows:

- First Generation
- Second Generation
- Third Generation
- Fourth Generation

First Generation (1945-1955)

Serial Processing

The evolution of operating systems began with serial processing. It marks the start of the development of electronic computing systems as alternatives to mechanical computers. Because of the flaws in mechanical computing devices, humans' calculation speed is limited, and they are prone to making mistakes. Because there is no operating system in this generation, the computer system is given instructions that must be carried out immediately. Programmers were incorporated into hardware components without using an operating system by the 1940s and 1950s. The challenges here are scheduling and setup time. The user logs in for machine time by wasting computational time. Setup time is required when loading the compiler, saving the compiled program, the source program, linking, and buffering. The process is restarted if an intermediate error occurs.

Example: Windows 95 and 98 are examples of serial processing operating systems.

Second Generation (1955-1965)

Batch System

The batched systems marked the **second generation** in the evolution of operating systems. In the second generation, the batch processing system was implemented,

which allows a job or task to be done in a series and then completed sequentially. The computer system in this generation does not have an operating system, although various operating system functionalities are available, such as FMS and IBSYS. It is used to improve computer utilization and application. On cards and tapes, jobs were scheduled and submitted. Then, using Job Control Language, they were successively executed on the monitors. The first computers employed in the batch operation method created a computer batch of jobs that never paused or stopped. The software is written on punch cards and then transferred to the tape's processing unit. When the computer finishes one job, it immediately moves on to the next item on the tape. Despite the fact that it is inconvenient for the users, it is designed to keep the pricey computer as busy as possible by running a leveraged stream of operations. Memory protection prevents the memory space that makes up the monitor from being changed, and the timer prevents the job from monopolizing the system. When the input and output devices are in use, the processor remains idle due to poor CPU utilization.

Example: MVS Operating System of IBM is an example of a batch processing operating system.

Third Generation (1965-1980)

Multi-Programmed Batched System

The evolution of operating systems embarks the **third generation** with multi-programmed batched systems. In the third generation, the operating system was designed to serve numerous users simultaneously. Interactive users can communicate with a computer via an online terminal, making the operating system multi-user and multiprogramming. It is used to execute several jobs that should be kept in the main memory. The processor determines which program to run through job scheduling algorithms.

Example: Windows and IOS are examples of multi-programmed batched operating systems.

Fourth Generation (1980-Now)

The operating system is employed in this age for computer networks where users are aware of the existence of computers connected to one another.

The era of networked computing has already begun, and users are comforted by a Graphical User Interface (GUI), which is an incredibly comfortable graphical computer interface. In the **fourth generation**, the time-sharing operating system and the Macintosh operating system came into existence.

1) Time-Sharing Operating System

The Time-sharing of operating systems had a great impact on the evolution of operating systems. Multiple users can access the system via terminals at the same time, and the processor's time is divided among them. Printing ports were required for programs having a command-line user interface, which required written responses to prompts or written commands. The interaction is scrolled down like a roll of paper. It was previously used to develop batch replacement systems. The user interfaces directly with the computer via printing ports, much like an electric

teletype. Few users shared the computer immediately, and each activity was completed in a fraction of a second before moving on to the next. By establishing iterations when they are receiving full attention, the fast server may act on a large number of users' processes at once. Multiple programs use time-sharing systems to apply to the computer system by sharing the system interactively.

Example: Unix Operating System is an example of a time-sharing OS.

1) **Macintosh Operating System**

It was based on decades of research into graphical operating systems and applications for personal computers. The photo depicts a Sutherland pioneer program sketchpad that was developed in 1960, employing many of the characteristics of today's graphical user interface, but the hardware components cost millions of dollars and took up a room. The initiative on massive computers and hardware improvements made the Macintosh commercially and economically viable after many research gaps. Many research laboratories are still working on research prototypes like sketchpads. It served as the foundation for anticipated products.

Example: Mac OS X 10.6.8 snow leopard and **OS X 10.7.5 Lion** are some examples of macintosh OS.

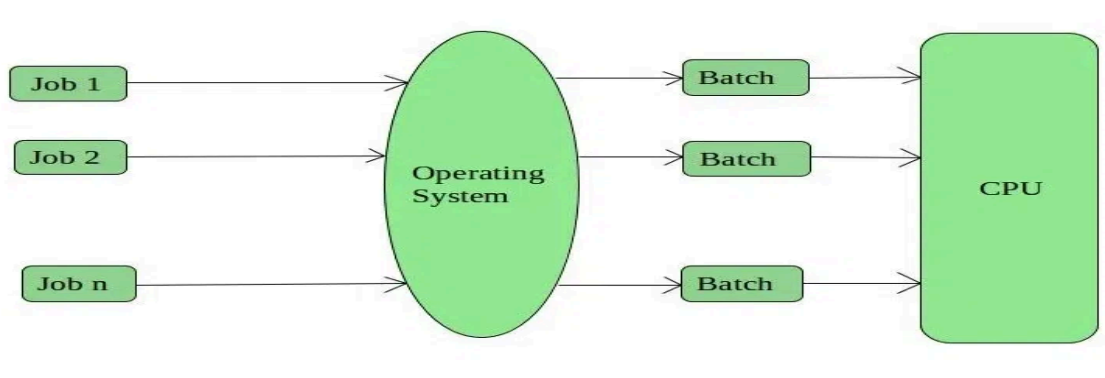
Types of Operating Systems

There are several types of Operating Systems which are mentioned below.

- Batch Operating System
- Multi-Programming System
- Multi-Processing System
- Multi-Tasking Operating System
- Time-Sharing Operating System
- Distributed Operating System
- Network Operating System
- Real-Time Operating System

1. Batch Operating System

This type of operating system does not interact with the computer directly. There is an operator which takes similar jobs having the same requirement and groups them into batches. It is the responsibility of the operator to sort jobs with similar needs.



Advantages of Batch Operating System

- It is very difficult to guess or know the time required for any job to complete. Processors of the batch systems know how long the job would be when it is in the queue.
- Multiple users can share the batch systems.
- The idle time for the batch system is very less.
- It is easy to manage large work repeatedly in batch systems.

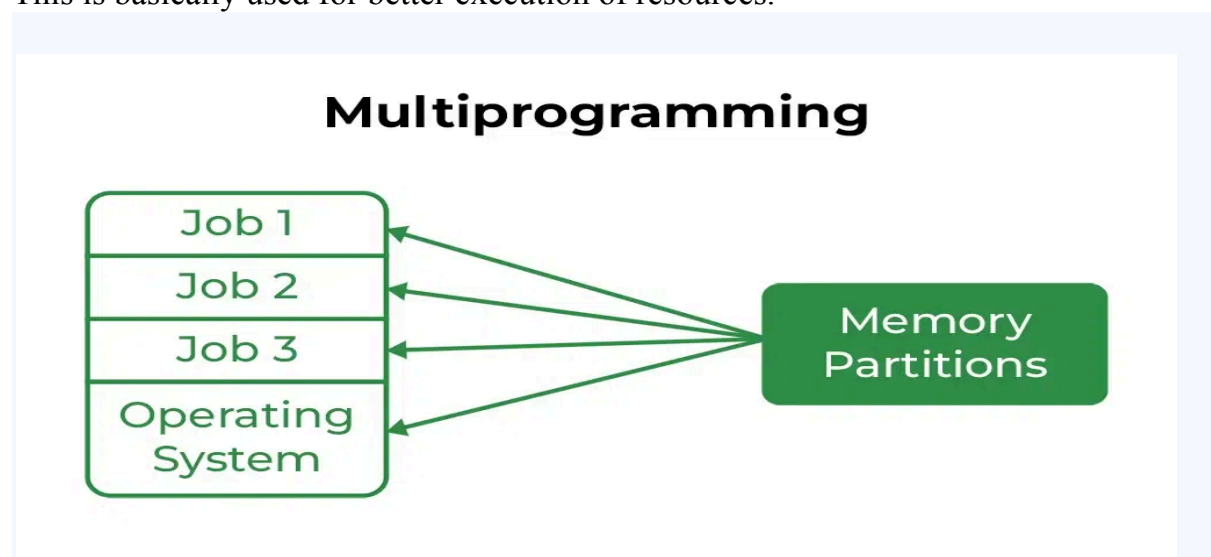
Disadvantages of Batch Operating System

- The computer operators should be well known with batch systems.
- Batch systems are hard to debug.
- It is sometimes costly.
- The other jobs will have to wait for an unknown time if any job fails.

Examples of Batch Operating Systems: Payroll Systems, Bank Statements, etc.

2. Multi-Programming Operating System

Multiprogramming Operating Systems can be simply illustrated as more than one program is present in the main memory and any one of them can be kept in execution. This is basically used for better execution of resources.



Advantages of Multi-Programming Operating System

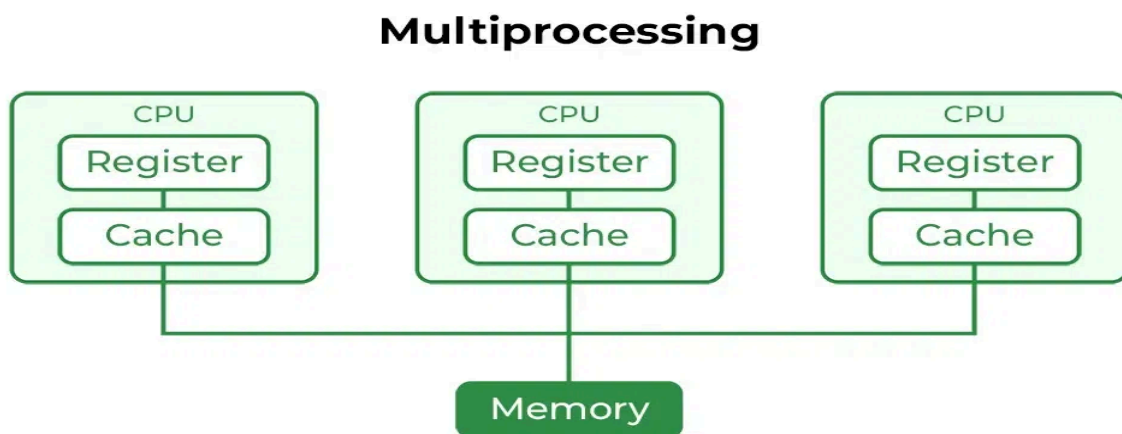
- Multi Programming increases the Throughput of the System.
- It helps in reducing the response time.

Disadvantages of Multi-Programming Operating System

- There is not any facility for user interaction of system resources with the system.

3. Multi-Processing Operating System

Multi-Processing Operating System is a type of Operating System in which more than one CPU is used for the execution of resources. It better the throughput of the System.



Advantages of Multi-Processing Operating System

- It increases the throughput of the system.
- As it has several processors, so, if one processor fails, we can proceed with another processor.

Disadvantages of Multi-Processing Operating System

- Due to the multiple CPU, it can be more complex and somehow difficult to understand.

4. Multi-Tasking Operating System

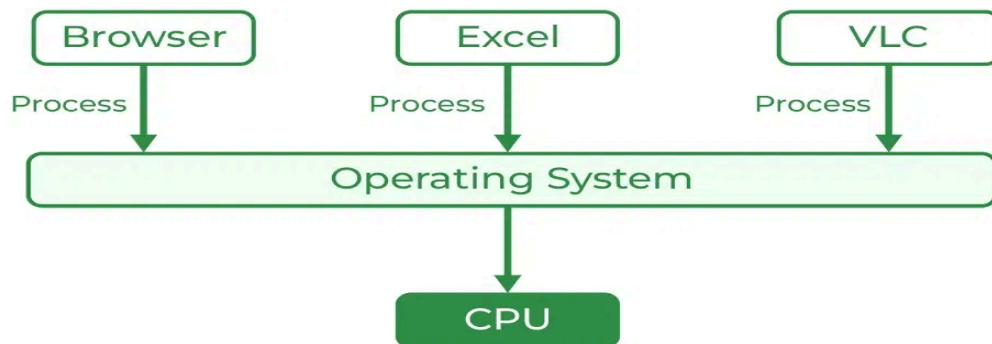
Multitasking Operating System is simply a multiprogramming Operating System with having facility of a Round-Robin Scheduling Algorithm. It can run multiple programs simultaneously.

There are two types of Multi-Tasking Systems which are listed below.

- Preemptive Multi-Tasking

- Cooperative Multi-Tasking

Multitasking



Advantages of Multi-Tasking Operating System

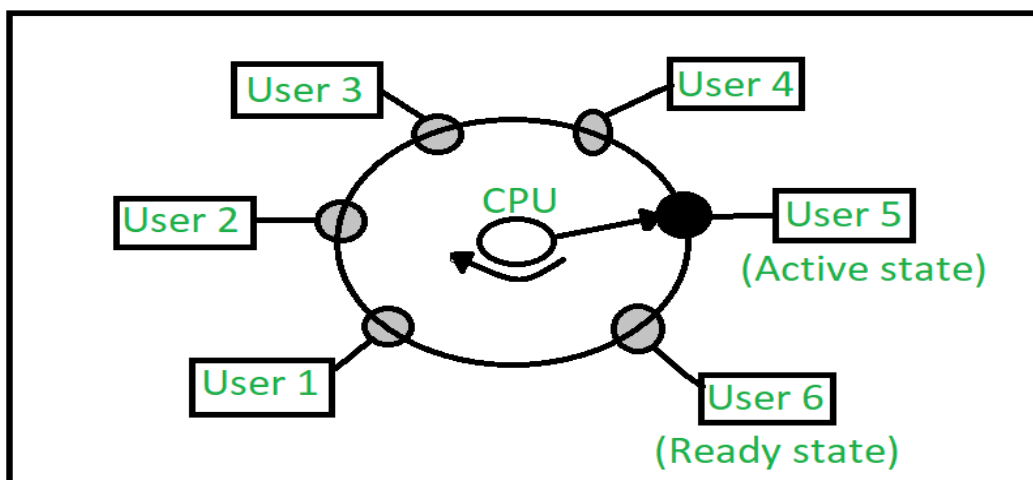
- Multiple Programs can be executed simultaneously in Multi-Tasking Operating System.
- It comes with proper memory management.

Disadvantages of Multi-Tasking Operating System

- The system gets heated in case of heavy programs multiple times.

5. Time-Sharing Operating Systems

Each task is given some time to execute so that all the tasks work smoothly. Each user gets the time of the CPU as they use a single system. These systems are also known as Multitasking Systems. The task can be from a single user or different users also. The time that each task gets to execute is called quantum. After this time interval is over OS switches over to the next task.



Advantages of Time-Sharing OS

- Each task gets an equal opportunity.
- Fewer chances of duplication of software.
- CPU idle time can be reduced.
- Resource Sharing: Time-sharing systems allow multiple users to share hardware resources such as the CPU, memory, and peripherals, reducing the cost of hardware and increasing efficiency.
- Improved Productivity: Time-sharing allows users to work concurrently, thereby reducing the waiting time for their turn to use the computer. This increased productivity translates to more work getting done in less time.
- Improved User Experience: Time-sharing provides an interactive environment that allows users to communicate with the computer in real time, providing a better user experience than batch processing.

Disadvantages of Time-Sharing OS

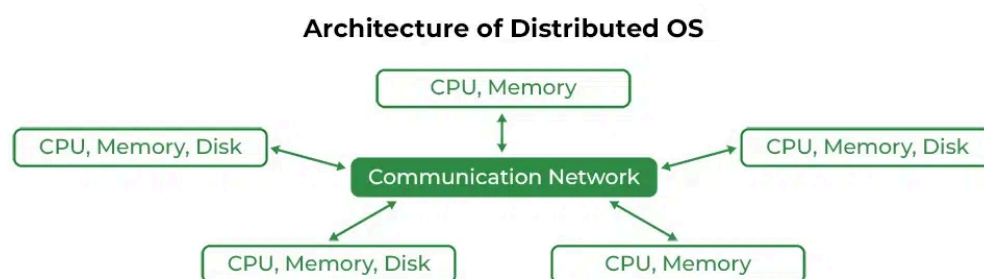
- Reliability problem.
- One must have to take care of the security and integrity of user programs and data.
- Data communication problem.
- High Overhead: Time-sharing systems have a higher overhead than other operating systems due to the need for scheduling, context switching, and other overheads that come with supporting multiple users.
- Complexity: Time-sharing systems are complex and require advanced software to manage multiple users simultaneously. This complexity increases the chance of bugs and errors.
- Security Risks: With multiple users sharing resources, the risk of security breaches increases. Time-sharing systems require careful management of user access, authentication, and authorization to ensure the security of data and software.

Examples of Time-Sharing OS with explanation

- **IBM VM/CMS:** IBM VM/CMS is a time-sharing operating system that was first introduced in 1972. It is still in use today, providing a virtual machine environment that allows multiple users to run their own instances of operating systems and applications.
- **TSO (Time Sharing Option):** TSO is a time-sharing operating system that was first introduced in the 1960s by IBM for the IBM System/360 mainframe computer. It allowed multiple users to access the same computer simultaneously, running their own applications.
- **Windows Terminal Services:** Windows Terminal Services is a time-sharing operating system that allows multiple users to access a Windows server remotely. Users can run their own applications and access shared resources, such as printers and network storage, in real-time.

6. Distributed Operating System

These types of operating system is a recent advancement in the world of computer technology and are being widely accepted all over the world and, that too, at a great pace. Various autonomous interconnected computers communicate with each other using a shared communication network. Independent systems possess their own memory unit and CPU. These are referred to as loosely coupled systems or distributed systems. These systems' processors differ in size and function. The major benefit of working with these types of the operating system is that it is always possible that one user can access the files or software which are not actually present on his system but some other system connected within this network i.e., remote access is enabled within the devices connected in that network.



Advantages of Distributed Operating System

- Failure of one will not affect the other network communication, as all systems are independent of each other.
- Electronic mail increases the data exchange speed.
- Since resources are being shared, computation is highly fast and durable.
- Load on host computer reduces.
- These systems are easily scalable as many systems can be easily added to the network.
- Delay in data processing reduces.

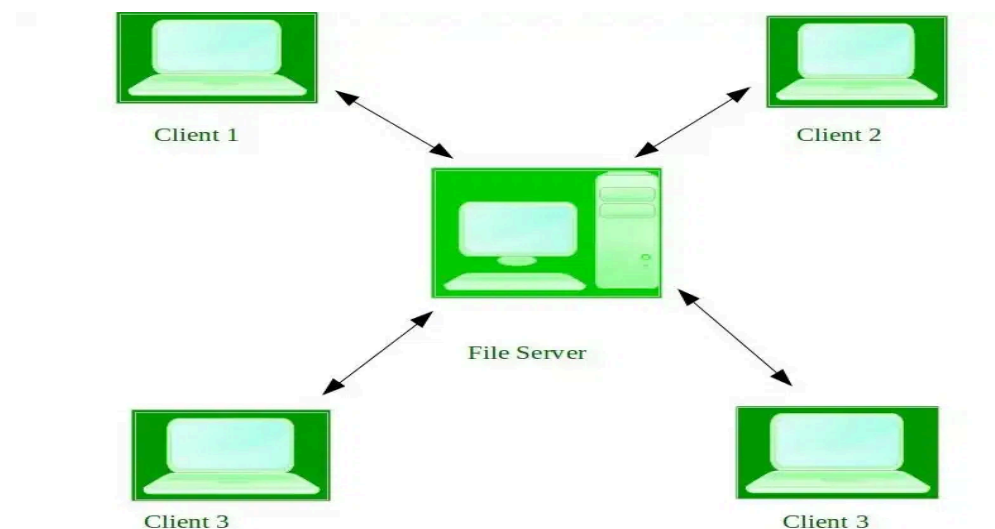
Disadvantages of Distributed Operating System

- Failure of the main network will stop the entire communication.
- To establish distributed systems the language is used not well-defined yet.
- These types of systems are not readily available as they are very expensive. Not only that the underlying software is highly complex and not understood well yet.

Examples of Distributed Operating Systems are LOCUS, etc.

7. Network Operating System

These systems run on a server and provide the capability to manage data, users, groups, security, applications, and other networking functions. These types of operating systems allow shared access to files, printers, security, applications, and other networking functions over a small private network. One more important aspect of Network Operating Systems is that all the users are well aware of the underlying configuration, of all other users within the network, their individual connections, etc. and that's why these computers are popularly known as tightly coupled systems.



Advantages of Network Operating System

- Highly stable centralized servers.
- Security concerns are handled through servers.
- New technologies and hardware up-gradation are easily integrated into the system.
- Server access is possible remotely from different locations and types of systems.

Disadvantages of Network Operating System

- Servers are costly.
- User has to depend on a central location for most operations.
- Maintenance and updates are required regularly.

Examples of Network Operating Systems are Microsoft Windows Server 2003, Microsoft Windows Server 2008, UNIX, Linux, Mac OS X, Novell NetWare, BSD, etc.

8. Real-Time Operating System

These types of OSs serve real-time systems. The time interval required to process and respond to inputs is very small. This time interval is called **response time**.

Real-time systems are used when there are time requirements that are very strict like missile systems, air traffic control systems, robots, etc.

Types of Real-Time Operating Systems

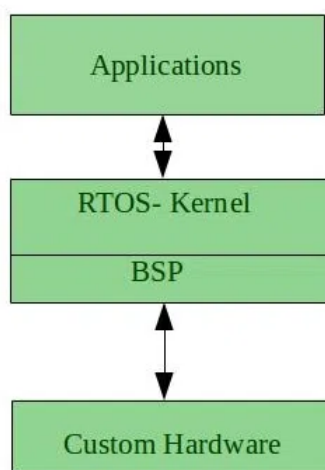
- **Hard Real-Time Systems:**

Hard Real-Time OSs are meant for applications where time constraints are very strict and even the shortest possible delay is not acceptable. These systems are built for saving life like automatic parachutes or airbags which are required to be readily available in case of an accident. Virtual memory is rarely found in these systems.

- **Soft Real-Time Systems:**

These OSs are for applications where time-constraint is less strict.

For more, refer to the [Difference Between Hard Real-Time OS and Soft Real-Time OS.](#)



Advantages of RTOS

- **Maximum Consumption:** Maximum utilization of devices and systems, thus more output from all the resources.
- **Task Shifting:** The time assigned for shifting tasks in these systems is very less. For example, in older systems, it takes about 10 microseconds in shifting from one task to another, and in the latest systems, it takes 3 microseconds.
- **Focus on Application:** Focus on running applications and less importance on applications that are in the queue.
- **Real-time operating system in the embedded system:** Since the size of programs is small, RTOS can also be used in embedded systems like in transport and others.
- **Error Free:** These types of systems are error-free.

- **Memory Allocation:** Memory allocation is best managed in these types of systems.

Disadvantages of RTOS

- **Limited Tasks:** Very few tasks run at the same time and their concentration is very less on a few applications to avoid errors.
- **Use heavy system resources:** Sometimes the system resources are not so good and they are expensive as well.
- **Complex Algorithms:** The algorithms are very complex and difficult for the designer to write on.
- **Device driver and interrupt signals:** It needs specific device drivers and interrupts signal to respond earliest to interrupts.
- **Thread Priority:** It is not good to set thread priority as these systems are very less prone to switching tasks.

Examples of Real-Time Operating Systems are Scientific experiments, medical imaging systems, industrial control systems, weapon systems, robots, air traffic control systems, etc.

Views of Operating System

An operating system is a framework that enables user application programs to interact with system hardware. The operating system does not perform any functions on its own, but it provides an atmosphere in which various apps and programs can do useful work. The operating system may be observed from the point of view of the user or the system, and it is known as the user view and the system view. In this article, you will learn the views of the operating system.

Viewpoints of Operating System

The operating system may be observed from the viewpoint of the user or the system. It is known as the user view and the system view. There are mainly two types of views of the operating system. These are as follows:

1. **User View**
2. **System View**

User View

The user view depends on the system interface that is used by the users. Some systems are designed for a single user to monopolize the resources to maximize the user's task. In these cases, the OS is designed primarily for ease of use, with little emphasis on quality and none on resource utilization.

The user viewpoint focuses on how the user interacts with the operating system through the usage of various application programs. In contrast, the system viewpoint focuses on how the hardware interacts with the operating system to complete various tasks.

1. Single User View Point

Most computer users use a monitor, keyboard, mouse, printer, and other accessories to operate their computer system. In some cases, the system is designed to maximize the output of a single user. As a result, more attention is laid on accessibility, and resource allocation is less important. These systems are much more designed for a single user experience and meet the needs of a single user, where the performance is not given focus as the multiple user systems.

2. Multiple User View Point

Another example of user views in which the importance of user experience and performance is given is when there is one mainframe computer and many users on their computers trying to interact with their kernels over the mainframe to each other. In such circumstances, memory allocation by the CPU must be done effectively to give a good user experience. The client-server architecture is another good example where many clients may interact through a remote server, and the same constraints of effective use of server resources may arise.

3. Handled User View Point

Moreover, the touchscreen era has given you the best handheld technology ever. Smartphones interact via wireless devices to perform numerous operations, but they're not as efficient as a computer interface, limiting their usefulness. However, their operating system is a great example of creating a device focused on the user's point of view.

4. Embedded System User View Point

Some systems, like embedded systems that lack a user point of view. The remote control used to turn **on** or **off** the tv is all part of an embedded system in which the electronic device communicates with another program where the user viewpoint is limited and allows the user to engage with the application.

System View

The OS may also be viewed as just a resource allocator. A computer system comprises various sources, such as hardware and software, which must be managed effectively. The operating system manages the resources, decides between competing demands, controls the program execution, etc. According to this point of view, the operating system's purpose is to maximize performance. The operating system is responsible for managing hardware resources and allocating them to programs and users to ensure maximum performance.

From the user point of view, we've discussed the numerous applications that require varying degrees of user participation. However, we are more concerned with how the hardware interacts with the operating system than with the user from a system viewpoint. The hardware and the operating system interact for a variety of reasons, including:

1. Resource Allocation

The hardware contains several resources like registers, caches, RAM, ROM, CPUs, I/O interaction, etc. These are all resources that the operating system needs when an application program demands them. Only the operating system can allocate resources, and it has used several tactics and strategies to maximize its processing and memory space. The operating system uses a variety of strategies to get the most out of the hardware resources, including paging, virtual memory, caching, and so on. These are very important in the case of various user viewpoints because inefficient resource allocation may affect the user viewpoint, causing the user system to lag or hang, reducing the user experience.

2. Control Program

The control program controls how input and output devices (hardware) interact with the operating system. The user may request an action that can only be done with I/O devices; in this case, the operating system must also have proper communication, control, detect, and handle such devices.

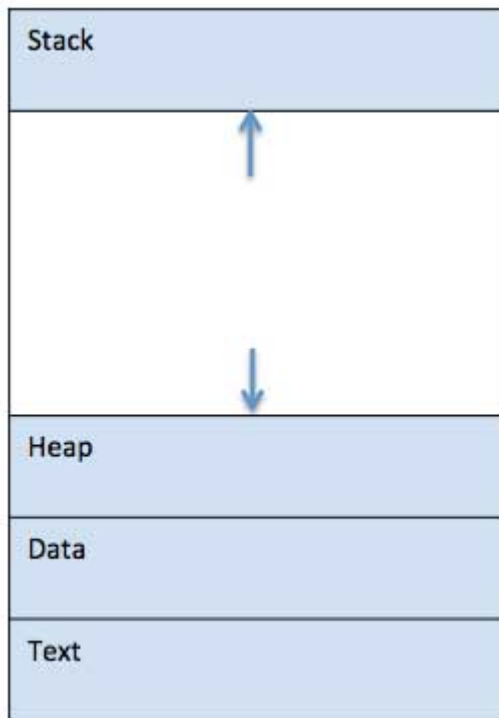
Process

A process is basically a program in execution. The execution of a process must progress in a sequential fashion.

A process is defined as an entity which represents the basic unit of work to be implemented in the system.

To put it in simple terms, we write our computer programs in a text file and when we execute this program, it becomes a process which performs all the tasks mentioned in the program.

When a program is loaded into the memory and it becomes a process, it can be divided into four sections — stack, heap, text and data. The following image shows a simplified layout of a process inside main memory –



S.N.	Component & Description
1	Stack The process Stack contains the temporary data such as method/function parameters, return address and local variables.
2	Heap This is dynamically allocated memory to a process during its run time.
3	Text This includes the current activity represented by the value of Program Counter and the contents of the processor's registers.
4	Data This section contains the global and static variables.

Services of Operating System

1. Program execution
2. Input Output Operations
3. Communication between Process
4. File Management
5. Memory Management
6. Process Management
7. Security and Privacy
8. Resource Management
9. User Interface
10. Networking
11. Error handling
12. Time Management

Program Execution

It is the Operating System that manages how a program is going to be executed. It loads the program into the memory after which it is executed. The order in which they are executed depends on the CPU Scheduling Algorithms. A few are FCFS, SJF, etc. When the program is in execution, the Operating System also handles deadlock i.e. no two processes come for execution at the same time. The Operating System is responsible for the smooth execution of both user and system programs. The Operating System utilizes various resources available for the efficient running of all types of functionalities.

Input Output Operations

Operating System manages the input-output operations and establishes communication between the user and device drivers. Device drivers are software that is associated with hardware that is being managed by the OS so that the sync between the devices works properly. It also provides access to input-output devices to a program when needed.

Communication between Processes

The Operating system manages the communication between processes. Communication between processes includes data transfer among them. If the processes are not on the same computer but connected through a computer network, then also their communication is managed by the Operating System itself.

File Management

The operating system helps in managing files also. If a program needs access to a file, it is the operating system that grants access. These permissions include read-only, read-write, etc. It also provides a platform for the user to create, and delete files. The Operating System is responsible for making decisions regarding the storage of all types of data or files, i.e, floppy disk/hard disk/pen drive, etc. The Operating System decides how the data should be manipulated and stored.

Memory Management

Let's understand memory management by OS in simple way. Imagine a cricket team with limited number of player . The team manager (OS) decide whether the upcoming player will be in playing 11 ,playing 15 or will not be included in team , based on his performance . In the same way, OS first check whether the upcoming program fulfil all requirement to get memory space or not ,if all things good, it checks how much memory space will be sufficient for program and then load the program into memory at certain location. And thus , it prevents program from using unnecessary memory.

Process Management

Let's understand the process management in unique way. Imagine, our kitchen stove as the (CPU) where all cooking(execution) is really happen and chef as the (OS) who uses kitchen-stove(CPU) to cook different dishes(program). The chef(OS) has to cook different dishes(programs) so he ensure that any particular dish(program) does not take long time(unnecessary time) and all dishes(programs) gets a chance to cooked(execution) .The chef(OS) basically scheduled time for all dishes(programs) to run kitchen(all the system) smoothly and thus cooked(execute) all the different dishes(programs) efficiently.

Secuirity and Privacy

- **Secuirity** : OS keep our computer safe from an unauthorised user by adding secuirity layer to it.Basically, Secuirity is nothing but just a layer of protection which protect computer from bad guys like viruses and hackers. OS provide us defenses like firewalls and anti-virus software and ensure good safety of computer and personal information.
- **Privacy** : OS give us facility to keep our essential information hidden like having a lock on our door, where only you can enter and other are not allowed . Basically , it respect our secrets and provide us facility to keep it safe.

Resource Management

System resources are shared between various processes. It is the Operating system that manages resource sharing. It also manages the CPU time among processes using CPU Scheduling Algorithms. It also helps in the memory management of the system. It also controls input-output devices. The OS also ensures the proper use of all the resources available by deciding which resource to be used by whom.

User Interface

User interface is essential and all operating systems provide it. Users either interface with the operating system through the command-line interface or graphical user interface or GUI. The command interpreter executes the next user-specified command.

A GUI offers the user a mouse-based window and menu system as an interface.

Networking

This service enables communication between devices on a network, such as connecting to the internet, sending and receiving data packets, and managing network connections.

Error Handling

The Operating System also handles the error occurring in the CPU, in Input-Output devices, etc. It also ensures that an error does not occur frequently and fixes the errors. It also prevents the process from coming to a deadlock. It also looks for any type of error or bugs that can occur while any task. The well-secured OS sometimes also acts as a countermeasure for preventing any sort of breach of the Computer System from any external source and probably handling them.

Time Management

Imagine traffic light as (OS), which indicates all the cars(programs) whether it should be stop(red)=>(simple queue) , start(yellow)=>(ready queue),move(green)=>(under execution) and this light (control) changes after a certain interval of time at each side of the road(computer system) so that the cars(program) from all side of road move smoothly without traffic.